

On the Number of Homogeneous Games Draft

M. Joris

Abstract

The result we prove, if correct, would be a pretty cool contribution to the combinatorics of homogeneous games, as we show that the number of homogeneous games is related to Fishburn numbers[8, 6]. These numbers show up in many unrelated-looking settings typically in things with partially constrained and ordered structures. Due to this, a plethora of consequences ranging from asymptotics, bijections with other objects, congruences, etc are now possible[2, 1, 4, 6]. The number of homogeneous games with n players is currently tabulated up to $n = 9$, namely the terms being 0, 1, 3, 8, 23, 76, 293, 1307, 6642, 37882 (given in OEIS A189359[7]). Our contribution makes it easy to compute h_n for any n , along with consequences such as

$$h_n \sim \left(\frac{6}{\pi^2}\right)^n n! \sqrt{n} \frac{12\sqrt{3}}{\pi^{5/2}} e^{\pi^2/12}$$

$$\frac{h_{n+1}}{nh_n} \rightarrow \frac{6}{\pi^2}$$

1 Statement of the result

Let h_n be the number of homogeneous games with n players, counted under the standard convention used in the enumeration of homogeneous games[3, 7]. We prove that for $n \geq 1$ we have

$$h_n = \sum_{j=1}^n F_j$$

where F_n are the Fishburn numbers[8], whose sequence is 1, 1, 2, 5, 15, 53, ... with generating function

$$F(x) = \sum_{n \geq 0} F_n x^n = \sum_{m \geq 0} \prod_{i=1}^m (1 - (1-x)^i)$$

written in the standard form[6]. Our proof uses work from Sudhölter, namely his recurrence for anti-step functions. Once the translation from homogeneous games to Sudhölter's functions is made, the rest of the argument is an ordinary generating-function calculation. The main result is the following

Theorem 1.1. *For every $n \geq 1$,*

$$h_n - h_{n-1} = F_n$$

Equivalently,

$$h_n = \sum_{j=1}^n F_j$$

In generating-function form,

$$\sum_{n \geq 1} h_n x^n = \frac{F(x) - 1}{1 - x}$$

2 Prerequisites and Sudhölter's functions

A weighted game is called homogeneous if it has a weighted representation in which every minimal winning coalition has total weight exactly equal to the quota. In other words, all minimal ways to win have the same total weight. A player is called a dummy if that player never changes the outcome of any coalition. Equivalently, adding a dummy player to a losing coalition never makes it winning. Dummies may be thought of as zero-weight players.

We write h_n for the number of homogeneous games with n players, under the standard equivalence convention, and g_n for the number of no-dummy homogeneous games with n players. Since every homogeneous game is obtained uniquely from a no-dummy homogeneous game by adjoining some number of final dummy players, Sudhölter's Lemma 1.1 gives[5]

$$h_n = \sum_{i=1}^n g_i$$

equivalently,

$$h_n - h_{n-1} = g_n$$

and thus it suffices to show that Theorem 1.1 is equivalent to proving

$$g_n = F_n$$

We use Sudhölter's enumeration of no-dummy homogeneous games, where he first reduces homogeneous games by removing dummies and one-person winning coalitions, then identifies the remaining games with incidence vectors and anti-step functions. His Lemma 1.1 gives the dummy deletion and the restoration of initial one-person-winning players, Lemma 1.4 gives the terminal-step deletion, Theorem 2.10 identifies homogeneous games without dummies and steps with incidence vectors, and Definition 3.1 identifies incidence vectors with anti-step functions[5].

Sudhölter then partitions the anti-step functions by type (k, p) in Definition 3.7[5]. We write $a_n^{k,p}$ for the number of n -person anti-step functions of type (k, p) , and

$$a_n^k := a_n^{k,0}$$

for the anti-step functions. The recurrence we use is stated below for these reduced functions.

For each $k \geq 1$,

$$A_k(x) = \sum_{n \geq 2k} a_n^k x^n$$

Sudhölter's formula gives[5]

$$a_n^{k,p} = a_{n-p}^k \binom{k+p-1}{p}$$

so summing over $p \geq 0$ gives the factor

$$\sum_{p \geq 0} \binom{k+p-1}{p} x^p = \frac{1}{(1-x)^k}$$

This is why in the next section we define

$$C_k(x) = \frac{A_k(x)}{(1-x)^k}$$

and

$$f(z) = \sum_{k \geq 1} C_k(x) z^k$$

so that

$$f(1) = \sum_{k \geq 1} C_k(x)$$

Thus $f(1)$ is the generating function on Sudhölter's anti-step side after the reduced counts have been expanded back to the full type (k, p) counts. There are two steps in translating this series to no-dummy homogeneous games. First, Sudhölter's terminal-step deletion, Lemma 1.4, shifts[5] the player number down by one. Thus a term counted by x^n in $f(1)$ first contributes to a no-dummy homogeneous game on $n - 1$ players. This accounts for the factor x^{-1} .

Second, the restoration of initial one-person-winning players, Lemma 1.1, says that we may adjoin any number of such players. If a game first contributes at size $n - 1$, then after adjoining $s \geq 0$ such players it contributes at sizes $n - 1, n, n + 1, \dots$ which contributes the geometric factor

$$1 + x + x^2 + \dots = \frac{1}{1 - x}$$

Hence a term counted by x^n in $f(1)$ contributes

$$x^{n-1}(1 + x + x^2 + \dots) = \frac{x^{n-1}}{1 - x}$$

to the no-dummy homogeneous-game generating function, and therefore, with

$$G(x) = \sum_{n \geq 1} g_n x^n$$

we get

$$G(x) = \frac{f(1)}{x(1 - x)} \tag{1}$$

Remark 2.1. $a_2^1 = 1$ is included here under the shift by one player and the restoration of initial one-person-winning players, it contributes $\frac{x}{1-x}$ to $G(x)$. This is the generating function for the 'exceptional no-dummy games' $(1; 1, \dots, 1)$, one in each positive number of players, though this is not needed further for our proof[5].

Remark 2.2. In Sudhölter's work and to tabulate the known values of h_n up until $n = 9$, incidence vectors are required for counting h_n , whereas with our method/proof this will not be needed[5].

3 The generating functions

We now introduce the generating functions attached to the reduced anti-step functions. For each $k \geq 1$, define

$$A_k(x) = \sum_{n \geq 2k} a_n^k x^n$$

The lower limit $n \geq 2k$ reflects the fact that $a_n^k = 0$ unless $n \geq 2k$. Define

$$C_k(x) = \frac{A_k(x)}{(1 - x)^k}$$

and

$$f(z) = \sum_{k \geq 1} C_k(x) z^k$$

The variable z is only for bookkeeping, and in particular,

$$f(1) = \sum_{k \geq 1} C_k(x)$$

The reason we want to find $f(1)$ is that, by (1),

$$\frac{f(1)}{x(1-x)} = \sum_{n \geq 1} g_n x^n$$

we get h_n through

$$h_n = \sum_{i=1}^n g_i$$

4 The recurrence

The reduced numbers $a_n^k = a_n^{k,0}$ satisfy $a_n^k = 0$ unless $k \leq \lfloor \frac{n}{2} \rfloor$, and following Sudhölter we set $a_2^1 = 1$ for technical convention, and $a_3^1 = 0$. Combining Sudhölter's Theorem 3.8 with Lemma 3.13 and Theorem 3.12 gives[5] the following recurrence for $n \geq 3$,

$$a_{n+1}^1 = \sum_{k=1}^{\lfloor n/2 \rfloor} \sum_{p=0}^{n-2k} \binom{k+p-1}{p} a_{n-p}^k \quad (2)$$

For $k \geq 2$, in the valid range

$$2 \leq k \leq \left\lfloor \frac{n+1}{2} \right\rfloor$$

the recurrence gives

$$a_{n+1}^k = \sum_{p=0}^{n-2k+2} \left[\sum_{j=k-1}^{\lfloor (n-p)/2 \rfloor} \binom{j}{k-1} \binom{j+p-1}{p} a_{n-p}^j \right] - a_n^{k-1} \quad (3)$$

So the goal is to turn these recurrences into a functional equation for $f(z)$ and solve that for $z = 1$.

5 From the recurrence to a functional equation

The $k = 1$ recurrence

Start with (2). Multiply both sides by x^{n+1} and sum over $n \geq 3$.

The left-hand side (I will just write 'LHS' from now on) is

$$\sum_{n \geq 3} a_{n+1}^1 x^{n+1}$$

Since

$$A_1(x) = \sum_{N \geq 2} a_N^1 x^N$$

the LHS sum is $A_1(x)$ with the terms $N = 2$ and $N = 3$ removed. Since $a_2^1 = 1$ and $a_3^1 = 0$, the LHS is simply

$$A_1(x) - x^2 - (0)x^3 = A_1(x) - x^2$$

Now let's consider the RHS. The recurrence is only valid for $n \geq 3$, but if we look at beginning the sum at $n = 2$, the expression becomes much cleaner

$$x \sum_{k \geq 1} \frac{A_k(x)}{(1-x)^k} = xf(1)$$

however for $n = 2$ we have included one term that does not belong to the recurrence, so we have to account for this. When $n = 2$, the only possible contribution is $k = 1$ and $p = 0$ and it contributes $x^3 a_2^1 = x^3$. Therefore the actual RHS, summed over $n \geq 3$, is

$$xf(1) - x^3$$

thus putting it together

$$A_1 - x^2 = xf(1) - x^3$$

Equivalently,

$$A_1 = x^2(1-x) + xf(1)$$

Since $A_1 = (1-x)C_1$, this gives

$$(1-x)C_1 = x^2(1-x) + xf(1) \tag{4}$$

The $k \geq 2$ recurrences

Now fix $k \geq 2$. We multiply (3) by x^{n+1} and sum over $n \geq 2k-1$ which is the natural range because the LHS is a_{n+1}^k , and terms with k first occur when $n+1 \geq 2k$, so therefore the LHS sums exactly to

$$A_k(x)$$

now look at the RHS. In the main double sum, put

$$m = n - p$$

The factor a_{n-p}^j becomes a_m^j and for fixed j , summing over $p \geq 0$ produces the binomial series

$$\sum_{p \geq 0} \binom{j+p-1}{p} x^p = \frac{1}{(1-x)^j}$$

and thus the unrestricted main contribution would be

$$x \sum_{j \geq k-1} \binom{j}{k-1} \frac{A_j(x)}{(1-x)^j} = x \sum_{j \geq k-1} \binom{j}{k-1} C_j$$

However, this unrestricted sum includes one boundary term that is not allowed by the original range $n \geq 2k - 1$ once again. The only forbidden case is when $j = k - 1$ and $m = 2k - 2$ and $p = 0$, which contributes $x^{2k-1}a_{2k-2}^{k-1}$. Now, lookin at the term $-a_n^{k-1}$, summing it over $n \geq 2k - 1$ gives

$$-x \sum_{n \geq 2k-1} a_n^{k-1} x^n$$

But

$$A_{k-1}(x) = \sum_{n \geq 2k-2} a_n^{k-1} x^n$$

So, quite nicely we get

$$-x \sum_{n \geq 2k-1} a_n^{k-1} x^n = -xA_{k-1}(x) + x^{2k-1}a_{2k-2}^{k-1}$$

i.e the same term appears once from the unrestricted main sum and once from the subtraction term, with opposite signs and cancel out. Hence no correction remains, and we get

$$A_k = x \sum_{j \geq k-1} \binom{j}{k-1} C_j - xA_{k-1}$$

Using $A_k = (1-x)^k C_k$, this becomes

$$(1-x)^k C_k = x \sum_{j \geq k-1} \binom{j}{k-1} C_j - x(1-x)^{k-1} C_{k-1} \quad (5)$$

Combining all $k \geq 2$

We now multiply (5) by z^{k-1} and sum over $k \geq 2$. (The purpose here of z is only to combine the infinitely many equations into one). The LHS becomes

$$\sum_{k \geq 2} (1-x)^k C_k z^{k-1} = \frac{f((1-x)z) - (1-x)C_1 z}{z}$$

with the subtraction term which becomes

$$-x \sum_{k \geq 2} (1-x)^{k-1} C_{k-1} z^{k-1} = -xf((1-x)z)$$

For the binomial term, write $r = k - 1$. Then $r \geq 1$, and

$$x \sum_{r \geq 1} z^r \sum_{j \geq r} \binom{j}{r} C_j = x \sum_{j \geq 1} C_j \sum_{r=1}^j \binom{j}{r} z^r$$

and since

$$\sum_{r=1}^j \binom{j}{r} z^r = (1+z)^j - 1$$

this is

$$x(f(1+z) - f(1))$$

and therefore

$$\frac{f((1-x)z) - (1-x)C_1z}{z} = x(f(1+z) - f(1)) - xf((1-x)z)$$

Now, using (4), namely

$$(1-x)C_1 = x^2(1-x) + xf(1)$$

the $f(1)$ terms cancel, and we obtain

$$(1+xz)f((1-x)z) = xzf(1+z) + x^2(1-x)z \quad (6)$$

6 Solving the functional equation

The functional equation

$$(1+xz)f((1-x)z) = xzf(1+z) + x^2(1-x)z$$

relates the value of f at $(1-x)z$ to the value of f at $1+z$. Since we ultimately need $f(1)$, we will choose z so that $(1-x)z = 1$, then repeat the same substitution again and again. To make this clean we can set $u = (1-x)z$, then $z = u/(1-x)$ and the functional equation becomes

$$f(u) = \frac{xu}{1-x+xu}f\left(1+\frac{u}{1-x}\right) + \frac{x^2(1-x)u}{1-x+xu} \quad (7)$$

Let's start with $u_0 = 1$, define

$$u_{r+1} = 1 + \frac{u_r}{1-x}$$

for our base case $r = 0$,

$$\frac{1 - (1-x)}{x} = 1$$

and then a direct induction gives

$$u_r = \frac{1 - (1-x)^{r+1}}{x(1-x)^r}$$

now assume the formula holds for r , then

$$u_{r+1} = 1 + \frac{1 - (1-x)^{r+1}}{x(1-x)^{r+1}} = \frac{x(1-x)^{r+1} + 1 - (1-x)^{r+1}}{x(1-x)^{r+1}} = \frac{1 - (1-x)^{r+2}}{x(1-x)^{r+1}}$$

and now we substitute $u = u_r$ into (7). Since

$$1 - x + xu_r = 1 - x + x \frac{1 - (1-x)^{r+1}}{x(1-x)^r} = 1 - x + \frac{1 - (1-x)^{r+1}}{(1-x)^r} = (1-x)^{-r}$$

we get

$$\frac{xu_r}{1-x+xu_r} = 1 - (1-x)^{r+1}$$

and then

$$\frac{x^2(1-x)u_r}{1-x+xu_r} = x(1-x)(1 - (1-x)^{r+1})$$

thus

$$f(u_r) = (1 - (1-x)^{r+1})f(u_{r+1}) + x(1-x)(1 - (1-x)^{r+1}) \quad (8)$$

therefore if we iterate (8) from $r = 0$ to $r = N - 1$, we get

$$f(1) = x(1-x) \sum_{m=1}^N \prod_{i=1}^m (1 - (1-x)^i) + \left(\prod_{i=1}^N (1 - (1-x)^i) \right) f(u_N)$$

Each factor $1 - (1-x)^i$ has zero constant term, so it is divisible by x . Therefore

$$\prod_{i=1}^N (1 - (1-x)^i)$$

is divisible by x^N , so the final term disappears as $N \rightarrow \infty$. Also Note $f(u_N)$ is an ordinary formal power series in x , with no negative powers of x . This follows from

$$u_N = \frac{1 - (1-x)^{N+1}}{x(1-x)^N} = (1-x)^{-N} (1 + (1-x) + \cdots + (1-x)^N)$$

which is a formal power series in x . Also

$$C_k(x) = \frac{A_k(x)}{(1-x)^k}$$

starts in degree at least $2k$, because $A_k(x)$ starts with x^{2k} . Hence

$$f(u_N) = \sum_{k \geq 1} C_k(x) u_N^k$$

is a well-defined formal power series. Therefore the remainder

$$\left(\prod_{i=1}^N (1 - (1-x)^i) \right) f(u_N)$$

is divisible by x^N . For any fixed coefficient of x^d , this remainder contributes nothing once $N > d$. Thus, as a formal power series, the remainder vanishes in the limit.

Letting $N \rightarrow \infty$, we obtain

$$f(1) = x(1-x) \sum_{m \geq 1} \prod_{i=1}^m (1 - (1-x)^i)$$

Equivalently,

$$1 + \frac{f(1)}{x(1-x)} = \sum_{m \geq 0} \prod_{i=1}^m (1 - (1-x)^i) \quad (9)$$

7 Proof of the main theorem

The RHS of (9) is the Fishburn generating function $F(x)$. Therefore

$$1 + \frac{f(1)}{x(1-x)} = F(x)$$

By (1),

$$G(x) = \frac{f(1)}{x(1-x)}$$

Hence

$$1 + G(x) = F(x)$$

or equivalently

$$G(x) = F(x) - 1$$

equating coefficients gives

$$g_n = F_n$$

for $n \geq 1$. Finally, since

$$h_n = \sum_{t=1}^n g_t$$

so

$$h_n - h_{n-1} = g_n = F_n$$

therefore

$$h_n - h_{n-1} = F_n$$

for $n \geq 1$, and equivalently

$$h_n = \sum_{j=1}^n F_j$$

which proves Theorem 1.1.

8 Consequences

These are immediate consequences/results from what I could find. These will just be stated without proof.

8.1 Congruences

One example,

$$h_{qn+r} \equiv h_{qn+r-1} \pmod{q}$$

whenever $F_{qn+r} \equiv 0 \pmod{q}$ [1, 4]

8.2 Asymptotics

$$h_n \sim \left(\frac{6}{\pi^2}\right)^n n! \sqrt{n} \frac{12\sqrt{3}}{\pi^{5/2}} e^{\pi^2/12}$$

and in particular,

$$\frac{h_{n+1}}{nh_n} \rightarrow \frac{6}{\pi^2}$$

[6]

8.3 Generating function

$$\sum_{n \geq 0} h_n x^n = \frac{1}{1-x} \left(\sum_{m \geq 0} \prod_{i=1}^m (1 - (1-x)^i) - 1 \right)$$

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